Closing-Line Lab: A Registered Football Forecasting Pipeline with Negative Historical Results and Sparse Betting Evidence

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Abstract

A market-anchored logistic blend achieves outcome log loss 1.007961 against 1.006958 for devigged closing prices on 3754 held-out football matches. The selected rule places only 2 historical paper bets: mean log closing-line value (CLV) is -0.026073 and profit is -0.832686 fictional units. Neither forecast accuracy nor these sparse bets establish an edge. We compare Dixon–Coles, Elo and market features using chronological fits over a privately cached dataset of 55476 rows. Selection is registered before the final-season evaluation, with two prospectively excluded schema-previewed fixtures. A full feature replay preserves the estimates within floating-point precision. Independent review identifies an invalid sparse-bootstrap significance rule; two synthetic-only replacements fail their calibration screen and confirmatory CLV inference is withdrawn before forward observations. A prospective pipeline records forecasts and near-kickoff quotes using hash chains and GitHub server receipts, with durable API-credit reservations. At the reporting snapshot there are 0 forward forecasts and 0 settled forward bets. This paper reports a negative historical comparison and an auditable prospective design, not completed forward evidence or executable trading performance.

1 Introduction

Football markets provide repeated three-outcome forecasts and an observable later price against which an earlier quote can be compared. This makes them useful for studying forecasting discipline, abstention and research provenance. A larger match count does not ensure independent observations, enough selected bets, or reliable inference. A strategy that largely reproduces market probabilities may forecast many matches while almost never crossing a meaningful value threshold.

Our question is whether a fixed model can obtain positive mean log CLV on decisions recorded before kickoff. The closing quote is a benchmark, rather than a known fair probability or proof of profitability. Our experiment is paper only. It creates no bookmaker account, order or transfer of money.

The contributions are: (i) a chronological comparison of a time-decayed Dixon–Coles model, Elo, and a market-anchored logistic blend, including every declared validation candidate and the negative held-out result; (ii) an auditable prospective protocol that binds quotes to pre-kickoff server receipts and reserves data credits remotely before transmission; and (iii) a documented withdrawal of unsupported inference after null simulation, with original registration tags and estimates preserved. The last contribution is a correction to the research procedure, not evidence that the forecasting model improved.

2 Related work

Dixon and Coles (1997) model football scores with dynamic team strengths and a correction for low-score dependence. We implement that structure with explicit regularization and a fixed fitting cadence; we do not claim to reproduce their empirical betting returns. Kaunitz et al. (2017) study strategies based on bookmakers' own prices and practical betting restrictions. Their work motivates a market-based comparison, but does not establish an edge for this dataset, selection rule or later market regime.

Bootstrap resampling originates in Efron (1979). Resampling match dates preserves within-date groups, but percentile endpoints from very few dates do not automatically have their nominal coverage. We demonstrate the failure for the original two-date rule and retain its endpoints as descriptive quantities. Bailey and López de Prado (2014) develop a Deflated Sharpe Ratio (DSR) correction for selection and non-normal returns. The repository's conservative trial accounting and variance floor are additional conventions; the resulting arithmetic cannot support precise inference from two nonzero return days.

3 Data

3.1 Historical source and access

The local dataset contains 55476 unique rows from 150 season–league CSV downloads from football-data.co.uk. It covers the top two divisions in England, Germany, Italy, Spain and France, from 2012/13 through the cached partial 2026/27 season. Not all rows have all odds fields or enter evaluation. We credit the provider and retain download URLs, byte counts and SHA-256 digests in a committed manifest. Downloads were sequential and cached once for the owner's private, non-commercial research under the documented owner decision. That decision is not an additional licence from the provider.

Raw CSVs, processed rows, historical predictions and individual historical bets remain local and ignored. Public artifacts consist of code, aggregate metrics/figures, manifests and fitted parameters. The notices distinguish this research use from unrestricted commercial or automated redistribution. Reproduction of exact historical rows requires the matching private cache; a mutable source URL cannot guarantee recovery of a lost snapshot.

3.2 Prices and timing assumptions

Historical decision prices are B365 non-closing fields; the primary later benchmark is AvgC. Non-closing fields are periodically captured snapshots, not verified opening odds. The CSVs do not supply auditable publication times for those prices. We therefore use a 24-hour-before-kickoff decision proxy and make only a conditional as-of claim about result features. Results with known kickoff times become available after 12 hours; date-only records after 36 hours. These assumptions do not reconstruct executable historical prices. Pinnacle historical fields are excluded because of the provider's feed-staleness warning.

Outcome scores use identical complete closing-price and known-result masks for model and benchmark. A corrected companion distinguishes zero missing closing triples from one missing result among the selected test predictions; the original result file mislabeled the combined exclusion. The frozen rule abstains on the unresolved match. The reusable simulator now keeps any selected unresolved stake as pending exposure rather than a loss.

3.3 Prospective source

The Odds API v4 supplies official programmatic event discovery, EU head-to-head quotes and scores (The Odds API, 2026). The forward population is EPL and Championship, chosen for coverage and budget before examining strategy returns. A scheduled workflow is configured hourly at minute 17 UTC. It attempts the first complete fresh decision quote 23–25 hours before kickoff and later a snapshot 15–75 minutes before kickoff. Quotes older than 30 minutes are rejected. Decisions prefer Pinnacle then Betfair Exchange EU; closes prefer Betfair Exchange EU then Pinnacle. No best-price search is performed.

Here "closing line" means the last captured pre-kickoff snapshot with a valid server receipt, not the literal last exchange price. Each observation records bookmaker, update and retrieval times, source hash and actual snapshot-to-kickoff gap. Delayed scheduling, missing coverage and budget restrictions can leave observations absent. Missing observations are never backfilled using post-match information.

4 Method

4.1 Forecasts

For decimal odds o_j, proportional de-vigging gives

$$q_j = \frac{o_j^{-1}}{\sum_{k\in\{H,D,A\}} o_k^{-1}}. \tag{1}$$

The market-only forecast uses these probabilities. Dixon–Coles estimates home and away goal rates from centered attack/defence effects, a common level and home advantage. Its weighted Poisson likelihood applies the four low-score adjustments of Dixon and Coles (1997). We use a 365-day half-life, a 1,095-day window, monthly refits, ridge penalty 0.02 and $\rho \in [-0.1, 0]$. Rates are clipped to $[e^{-2}, e^{2}]$ and outcome probabilities normalize a 0–30 goal grid. Fits require at least 100 earlier league matches; optimizer failure stops evaluation.

Elo starts at 1,500 per team/league, uses $K = 20$, home advantage 75, and Davidson-style three-way probabilities proportional to $(r, 0.7\sqrt{r}, 1)$, where $r = 10^{(R_H+75-R_A)/400}$. Updates use only results available before the decision. The blend uses shared residual weights and outcome intercepts around the market anchor:

$$p_j \propto \exp[\log q_j + b_j + w_{DC}(\log p_{DC,j} - \log q_j) + w_E(\log p_{E,j} - \log q_j)]\,. \tag{2}$$

The two shared weights and three intercepts have bounds $[-1, 1]$ and ridge penalty 10. A July fit uses completed earlier seasons only, with at least 1,000 feature rows. There is no current-season coefficient optimization.

Historical DC/Elo features update under the fixed chronological rules. Forward parameters are exported once after the original registration and test evaluation: DC/Elo use results available at export, while blend fitting labels end before the test season. All parameters then stay fixed. This deliberately introduces cadence and price-source shifts; hosted jobs never download football-data rows or retrain the model.

4.2 Paper selection, account and metrics

Choose at most one side maximizing $p_j\tilde{o}_j - 1$, ties in H/D/A order, with odds in $[1.2, 10]$ and edge at least 0.05. Exchange effective odds are $\tilde{o} = 1 + 0.98(o - 1)$; book odds are unchanged. The

account begins at 100 fictional units. Stake fraction is $\min(0.01, 0.25(p_j\tilde{o}_j - 1)/(\tilde{o}_j - 1))$, subject to available cash and a 10% open-exposure cap. Open stakes are valued at cost; future winnings cannot fund simultaneous decisions. Unknown outcomes remain pending and are excluded from realized profit, ROI and the settled-bet CLV summary without removing them before selection.

Primary log CLV is $\log(o_{\text{taken}}/o_{\text{close}})$ on the selected side, unweighted across settled bets with an observed close. Secondary checks use $\log(o_{\text{taken}}q_{\text{close}})$ and same-book raw CLV where available. Raw cross-book CLV can reflect margin differences. Outcome log loss and summed multiclass Brier score compare forecasts with de-vigged closing probabilities on the same matches. Closing-distribution cross entropy is a separate market-distance diagnostic. Neither paper P&L nor displayed quotes establish executable fills, depth or attainable returns.

5 Protocol

5.1 Splits, registration and trial accounting

Features begin in 2015/16 after initial history. Validation spans 2022/23–2024/25; the final full season 2025/26 is evaluated after the original `forward-v1` tag. Four forecast families crossed with minimum edges 0.03/0.05 form eight declared candidates. Selection minimizes validation outcome log loss, with exact ties choosing the higher threshold. Thresholds do not change forecast probabilities; ties therefore do not select on CLV or profit. All candidates are retained. No league, side or lead-time search is used.

Two test fixtures had been displayed for schema inspection before fitting and were prospectively excluded from primary test metrics. Thus the full season was not fully blinded. Earlier test results may enter later test features under the fixed as-of rule; aggregate test metrics cannot guide model or threshold changes. The registered blend and parameter file are unchanged by review corrections.

Every strategy candidate and documented iteration, including failures, is appended to a trial registry. The original validation and test DSR diagnostics retain their evaluation-time counts rather than acquiring later attempts retrospectively. Subsequent forward reports use the growing registry. Counting operational attempts is conservative bookkeeping, not an estimate of independent strategy searches or formal familywise-error control. No confirmatory family of model-comparison tests, Holm correction or superiority claim is made.

The original tag and subsequent prospective tags remain in history. The active `forward-v1.3` amendment repairs closing evidence, end-window capture, pending settlements and durable credit accounting, and explicitly withdraws the original significance rule before any forward forecast. Original validation/test artifacts are preserved; corrected results are a separately labeled replay companion.

5.2 Sparse inference correction

The original protocol treated a positive lower 95% date-bootstrap percentile endpoint as evidence of positive CLV. With two independent symmetric continuous zero-mean observations, the resampled support lies between those two observations. The lower percentile is positive when both are positive, which occurs with probability 0.25. Coverage of zero is only one half. Nominal percentile labels therefore gave misleading inference.

We retain the resampling endpoints as descriptive quantiles. Two replacement candidates were screened using synthetic data only: bet-weighted weekly means, Bartlett HAC variance with four lags, a Student-t critical value, unequal week sizes, and increasing minimum information. The null grid combines normal, Student-$t(5)$ and negative centered lognormal innovations, weekly AR(1)

coefficients 0, 0.3 and 0.6, and 30, 60 or 120 populated weeks. Each scenario uses 10,000 replications with seed 20261029 and a burn-in. Acceptance requires every scenario's Wilson 95% upper false-positive bound to be at most 0.05. This empirical screen would not establish universal validity even if passed.

Both candidates fail (Table 4). After two attempts we stop inferential tuning. For this pilot, below 100 bets or 60 populated weeks CLV inference returns `INSUFFICIENT_INFORMATION`; otherwise it returns `INFERENCE_NOT_VALIDATED`. It always supplies no inferential interval and no positive-evidence decision. The primary estimand remains mean CLV, but a future confirmatory study requires a separately validated preregistration. Secondary DSR is `NOT_ESTIMABLE` below 30 nonzero return days or 60 calendar days. Its old arithmetic remains in artifacts, with assumptions and bespoke conventions disclosed.

5.3 Forward inclusion and publication

The planned decision window is 29 October 2026 through 29 January 2027 exclusive (UTC). Inclusion is by decision time: existing forecasts can receive closes and settlement after the window ends. There is no performance-based stopping or extension.

Forecasts and closing snapshots append to a SHA-256 chain. Publication checks a kickoff buffer, pushes without rewriting history, and dispatches an audit. A GitHub run's server creation time and head SHA must bind the exact ledger content before kickoff; local git author timestamps are insufficient. Settlement must reference the latest receipted close and replay its prices, bookmaker, gap and raw/secondary CLV. Without such evidence every close-derived field is null, although a known outcome may settle P&L.

A separate remote branch records a compare-and-swap credit reservation before each paid HTTP call. Ambiguous failure leaves a pending reservation and blocks paid requests even on a new runner or after a provider reset. Manual reconciliation retains ambiguous charges; paid calls are not retried. The 400-credit calendar-month cap and 100-credit provider reserve leave headroom on the intended free plan. Free fixture discovery and due-window gating reduce requests. Actual coverage is not guaranteed.

6 Results

6.1 Historical forecast and selection results

Table 1 includes every declared validation strategy. The selected blend is nearer to closing log loss than DC or Elo, but remains worse than the closing baseline. The higher threshold yields only 4 validation bets. This selection rule is fixed before inspecting the final-season aggregate and is retained despite weak betting power.

Table 2 shows paired forecast scores. On 3754 test matches, the blend's log-loss gap to closing is 0.001003 nats, a deterioration. Brier scores also favor closing at their point estimates. Calibration bins in Figure 1 are descriptive and do not establish betting value.

6.2 Selected bets, replay and power

The 2 selected test bets both lose. Their mean log CLV is -0.026073 and paper profit is -0.832686 units (Table 3). ROI is on staked capital, not the entire initial account. The two-date bootstrap endpoints are the individual CLV extremes; they are not calibrated confidence limits. DSR is not estimable with 2 nonzero return days despite 306 calendar observations.

Model	Edge	Log loss	Bets	Mean log CLV
Market	0.03	1.006368	0	—
Market	0.05	1.006368	0	—
Dixon–Coles	0.03	1.025917	3401	-0.004311
Dixon–Coles	0.05	1.025917	3192	-0.004204
Elo	0.03	1.025936	3465	-0.007975
Elo	0.05	1.025936	3272	-0.007094
Blend	0.03	1.005764	14	-0.048238
Blend	0.05	1.005764	4	-0.039889

Table 1: All declared validation candidates on the common 11486 scored matches. An undefined CLV mean is shown as a dash. No threshold is chosen on CLV.

Population / forecast	Matches	Log loss	Brier
Validation: blend	11486	1.005764	0.601869
Validation: close	11486	1.002834	0.599915
Test: blend	3754	1.007961	0.603416
Test: close	3754	1.006958	0.602787

Table 2: Selected blend and proportional de-vigged closing benchmark on identical masks. Lower log loss and summed multiclass Brier are better.

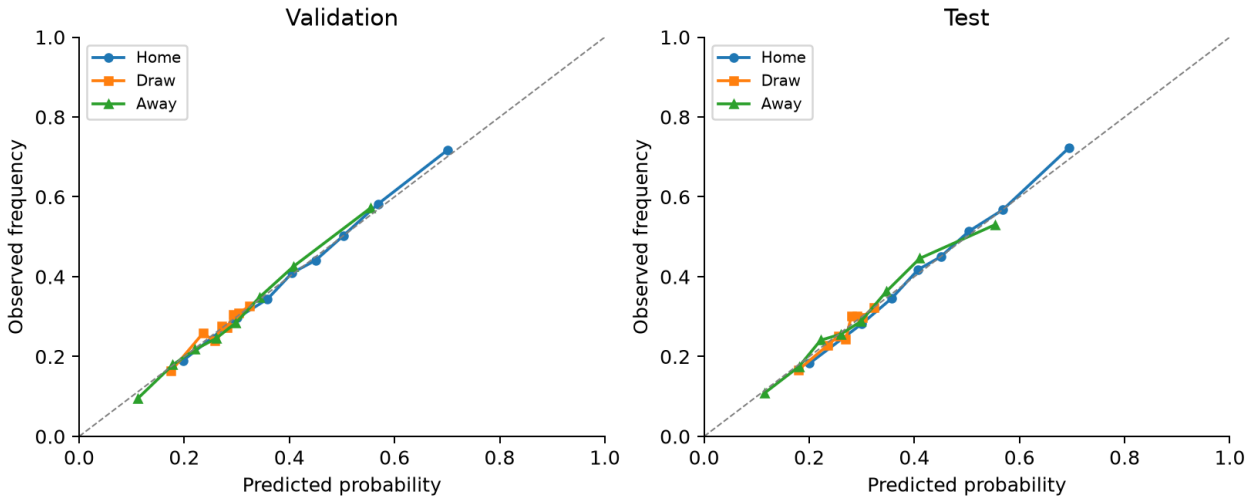


Figure 1: H/D/A calibration from committed aggregate quantile bins. Curves show mean predicted probability against observed frequency, without inferential error bars.

Selected-rule quantity	Validation	Test
Settled bets	4	2
Pending bets	0	0
Mean log CLV	-0.039889	-0.026073
Bootstrap 2.5% quantile	-0.086783	-0.159849
Bootstrap 97.5% quantile	0.016521	0.107702
Paper profit (units)	2.306993	-0.832686
ROI on settled stakes	0.970305	-1.000000
Nonzero return days	4	2
DSR inference	Not estimable	Not estimable

Table 3: Selected-rule betting summaries. Bootstrap endpoints are descriptive quantiles from date resampling, not reliable confidence limits. Fictional units have no cash value.

Full feature regeneration and isolated replay preserve every original headline value across all eight validation rules and the selected test rule. The maximum saved-prediction probability deviation is 1.11e-16. Frozen reports, selection and parameter bytes remain unchanged. The correction companion separates price/result missingness and adds pending exposure and sparse-inference statuses.

A normal-approximation planning calculation based on only 4 validation bets suggests 23821 bets for 10 log-CLV basis points at one-sided 5% size and 80% power. This is an unstable variance extrapolation, not a reliable sample-size requirement or a validated test design. Two test bets cannot estimate planning power. The proposed three-month window is unlikely to identify a small edge for such a sparse rule.

6.3 Null checks and prospective status

Table 4 and Figure 2 retain both failed inference candidates. The larger critical parameter reduces rejection, but its worst upper bound remains 0.052782, above the prespecified screen. Neither setting is a validated confidence level. No further outcome-driven or synthetic inferential tuning is performed.

Candidate	Min. weeks	Critical parameter	Max. FPR	Upper bound	Screen
1	30	0.990	0.126400	0.133057	FAIL
2	60	0.999	0.048400	0.052782	FAIL

Table 4: Synthetic-only inference screen. FPR is rejection under the zero-mean null. "Critical parameter" specifies the candidate's nominal interval calculation, not achieved coverage. Both candidates also require 100 bets. The upper bound is Wilson 95% per scenario.

At the committed reporting snapshot there are 0 forward forecasts, 0 selected paper bets and 0 settled forward bets. Successful scheduled executions observed: 0. Manual workflow success and an empty-ledger audit verify deployment paths, not an on-time live forecast, close or settlement. Backtest versus forward performance is therefore not yet comparable. There is no claim of demonstrated prospective alpha or a completed forward experiment.

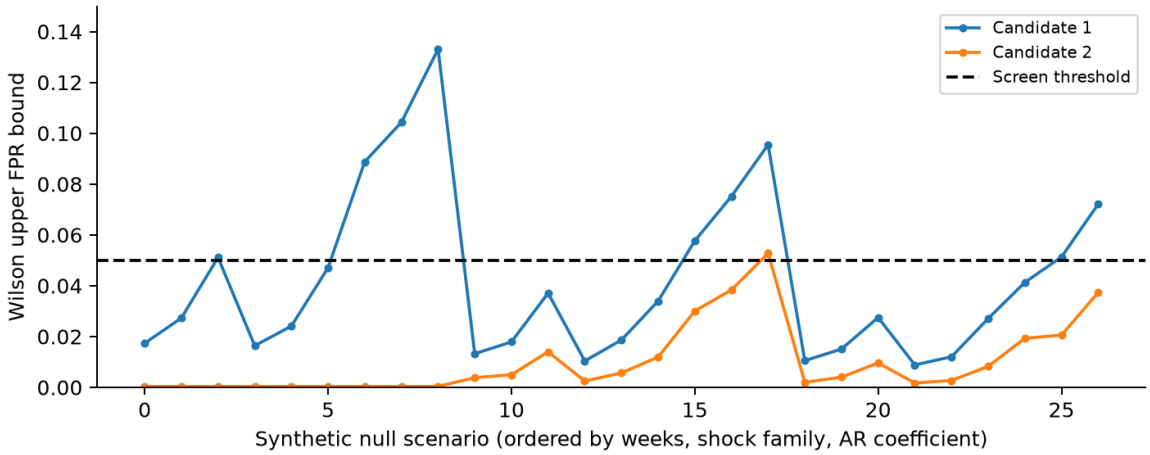


Figure 2: Upper false-positive bounds across the complete synthetic scenario grid. Ineligible sample sizes never reject. Every scenario must satisfy the dashed threshold.

7 Limitations

Historical prices lack trustworthy as-of publication times. Result-availability proxies, cross-book margins and a closing market that need not be efficient constrain interpretation. The test season contains declared schema-preview exceptions. Data reuse across chronological features is intentional, but does not create independent match observations. Abstention leaves the selected strategy with too few bets for confirmatory inference.

Forward parameters remain fixed while historical DC/Elo updated; source and cadence shifts may dominate a future comparison. Hourly workflows can be delayed or dropped. The observed close can precede the literal final line substantially; source update gaps are reported separately. API quotas, quote completeness, name mappings and outages affect inclusion. The scores endpoint's limited lookback can leave outages unresolved. Missingness may be systematic, so complete-case CLV need not describe all planned decisions.

Indicative quotes, a fixed exchange commission and fictional stakes do not model depth, limits, slippage or account restrictions. Neither CLV nor paper returns establish attainable profit. Hash chains and server receipts are tamper-evident against retained history, not immutable: repository owners can delete history and runs. Private evidence requires access. The public aggregate artifacts do not permit exact row-level replay without the private source cache. Synthetic null scenarios are finite and cannot prove coverage under arbitrary heavy tails or dependence; disabling confirmation is the supported decision here.

8 Reproducibility

The original pre-test registration is commit `7c16ca0b79f71a30f7820e26289d651d22ecbb58`; active prospective correction v1.3 is `8b59f2a755a99fc3bc8399c3ef944002a0681cb8`. The full-feature correction replay uses source `8b59f2a755a99fc3bc8399c3ef944002a0681cb8`. Tables, numerical macros and figures are generated by `paper/make_tables.py` from Git-committed aggregate artifacts at `1e5fd9d421686734018d1a4cefb88373051f32a3`. Their SHA-256 digests and the generator hash are saved in `paper/generated/provenance.json`. Displayed decimals use a consistent rounding-to-nearest rule; machine-readable artifacts retain full precision.

Python 3.12 dependencies are locked with uv. Tests cover as-of features, de-vigging, CLV, ledger evidence, budget loss across fresh runners, pending stakes and end-window closing. The recorded engineering snapshot passes 59 tests. Commands from the repository root are:

```
uv sync --frozen
make check test
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 uv run cll replay \
  --output-dir runs/reviewer-replay --rebuild-features
uv run python scripts/calibrate_inference.py --min-weeks 30 \
  --critical-coverage 0.99 --output runs/candidate1.json
uv run python scripts/calibrate_inference.py --output runs/candidate2.json
make verify
uv run python paper/make_tables.py
tectonic -X compile paper/main.tex
```

The replay command needs the authorized local cache and writes isolated ignored outputs; it does not rerun selection or overwrite registered artifacts. Null simulations need no market data. The paper generator reads committed blobs directly, never uncommitted local rows. Reference metadata and identifiers were resolved through Crossref, the arXiv API or official documentation and retained in `paper/reference_verification.json`.

9 Conclusion

The selected market-anchored blend does not beat the closing benchmark at held-out point estimates. Its sparse selected bets show negative CLV and losses but cannot support precise inference. Review corrections preserve those estimates while repairing settlement, publication evidence and durable budget accounting. Two attempted null-calibrated inference replacements fail; the pilot accordingly remains descriptive. The prospective infrastructure and retained provenance support a future evaluation, but no operating forward edge has yet been demonstrated.

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